

Project Design Phase

Proposed Solution Template

Date	13 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Prospective home buyers, sellers, and real estate analysts often struggle to understand what actually drives house prices in a given market. Raw housing data (prices, square footage, bedrooms, bathrooms, condition, renovation history, location) is available but not easily interpretable without technical tools, making it hard for non-technical stakeholders to identify pricing patterns, the impact of renovations, or how house age and features relate to sale value.
2.	Idea / Solution description	We built an interactive Tableau dashboard that visualizes ~21,600 records of King County-style housing data. The solution cleans and prepares the raw dataset, then presents it through four coordinated visualizations — a KPI summary table, a renovation-based sales histogram, a house-age vs. renovation-status pie chart, and a bedroom/bathroom/floor-based age distribution chart — combined into a single 2x2 dashboard and supported by a guided storyboard that walks viewers through key insights step by step.
3.	Novelty / Uniqueness	Unlike static spreadsheets or one-off charts, this solution combines multiple related dimensions (renovation history, house age, room configuration, basement area) into one cohesive, interactive dashboard rather than isolated reports. The storyboard layer adds a guided narrative on top of the raw dashboard, making the insights accessible to non-technical viewers without requiring them to interpret each chart independently.

4.	Social Impact / Customer Satisfaction	The dashboard helps home buyers and sellers make more informed pricing decisions by visually showing how renovation status and house features affect value, reducing reliance on guesswork or opaque listing prices. Real estate analysts get a quick, shareable reference tool instead of having to rebuild analysis from raw data each time.
5.	Business Model (Revenue Model)	As an educational/internship project, the dashboard itself is published free via Tableau Public. In a real-world extension, this could support a freemium model — a free public dashboard for general trends, with a premium version offering zipcode-level drill-downs, predictive price estimates, or API access for real estate agencies and investors.
6.	Scalability of the Solution	The Tableau workbook is modular — additional sheets, filters, or an updated/larger dataset can be incorporated without redesigning the existing dashboard structure. The same architecture could extend to other counties/regions or be connected to a live data source (e.g., MLS feed) instead of a static CSV for real-time updates.